Football Highlights Detection using Deep Learning: A Comparative Study of Temporal Architectures

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Abstract

Highlight reel generation on full-length video footage of football games, regardless of whether it is automatic, is a challenging temporal action localization problem due to severe imbalance between classes, the long-tailed distribution of salient events, and the high cost of dense annotation. We describe an end-to-end pipeline that recognizes three event types of key matches that are Goals, Cards, and Substitutions, against a dominant Background class, and assembles the most informative parts of these matches into a compact highlight reel. Similar to the SoccerNet feature pipeline, the sampled frames (2 frames per second) are encoded with a pre-trained ResNet-152 and projected to 512 dimensions using Principal Component Analysis (PCA). Three temporal classifiers are trained on a 60-match subset of SoccerNet-v2 and rigorously compared: fully connected baseline, a Bidirectional Long Short-Term Memory (BiLSTM) network, and a Transformer encoder. Highlight selection can then be modeled as a 0/1 Knapsack problem with a user-specified time budget. The CNN+BiLSTM has the best detection performance with a test accuracy of 96.3% and a mean Average Precison (mAP) of 35.1% which is approximately half of the state-of-the-art detection performance using full data. Transformer-based temporal modeling is data-hungry (achieves similar accuracy, 93.9%), but with a much lower 8.9% mAP, which is indicative of data-hungry models that tend to collapse to the majority Background classification under our resource constraints. Our results support the idea that recurring inductive biases are very competitive in fine-grained temporal action spotting in sports video when in the small-data regime.

Keywords- action spotting, football highlights, BiLSTM, Transformer, ResNet-152, SoccerNet, temporal modeling, knapsack optimization, video summarization.

I. INTRODUCTION

Football is also one of the most widely consumed broadcast media in the world generating tens of thousands of hours of new video each season in professional leagues, continental competitions and amateur tournaments. But of this film one can truly watch only a small part of it in the full-length raw form. The most common form of consumption is the highlight reel, a edited three to ten-minute compaction of a 90-minute match that still retains the goals, the disciplinary moments, the substitutions, and the most spectacular moments of play. The scale production of these reels has become a key logistical bottleneck of broadcasters, streaming platforms, and statistical-analysis providers, and it is this production bottleneck that drives an automated pipeline.

The traditional production workflow is manually based. A human operator scrubs ninety to one hundred and twenty minutes of footage at various playback rates, marks in- and out-points on a non-linear editor, and exports a final cut. This process is labor-intensive even with elite matches, where dedicated production teams can keep up with live broadcasts; it becomes economically unsustainable among the long tail of lower-division, women, youth, and amateur football, which all contribute to the majority of matches played and only a tiny fraction of the edited content. Even partial automation of this process would open up the production of highlights to these underserved leagues, lower the operating cost of major broadcasters, and provide near-real-time reels which are synchronized with the actual live event.

In the research viewpoint, the automatic highlight generation is considered a special case of the temporal action localization in long and untrimmed video, which is a problem that is still much harder to solve than its counterpart that is trimmed into clips. Broadcast football is especially challenging due to three of its properties. To begin with, the events of interest comprise less than five percent of the entire timeline, creating a severely imbalanced classification problem in which an insignificant majority-class predictor already achieves high accuracy. Second, the visual context immediately preceding and following a salient event are almost identical to everyday play; an approach to a goal looks virtually the same as a routine attacking move that fizzles out, and the model must be able to utilize subtle cues that are spaced over a series of seconds instead of being on a single discriminative frame. Third, the misclassification cost is asymmetric: you can miss a goal at the cost of editorial disapproval; but you can add a few additional seconds of buildup at the cost of editorial disapproval.

The uses of a sound automated system are far greater than editorial broadcast. The same per-frame event detector can be reused in sports analytics to fill match-event databases, to drive real-time win-probability models, or to automatically label training video to help coaches. Within the consumer space, highlight generators are being progressively directly integrated into streaming platforms, enabling viewers to jump

to the next meaningful. Similar systems are used by federations and clubs to create social-media-ready clips minutes after a match has concluded. The technical heart of all these applications - a model which maps long video into a thin stream of typed event timestamps - is the same problem we solve in this paper.

Recent advances in deep learning to understand video have focused primarily on action recognition of short, trimmed video clips. Spotting of temporal action in long, uncut broadcasts is a relatively unexplored problem, and the SoccerNet benchmark [1], [2] has become the standard of evaluation in the football context in particular. State of the art entries on the full benchmark report achieve over 70 Average Precision (mAP) on hundreds of matches, multi-GPU clusters and features that are pre-extracted and published with the benchmark. The practical question of what can be done with the constraints of a single free-tier Google Colab GPU and about 12% of the dataset has received far less attention, yet it is this regime that determines whether the technology can be accessible to academic groups, undergraduate research projects and practitioners outside well-funded broadcast laboratories.

That is the question we answer in this work by a controlled comparison of three temporal architectures superimposed on top of a shared spatial encoder. Each frame is then coded by a pre-trained ResNet-152 [3] and reduced to 512 dimensions using Principal Component Analysis, which matches the official SoccerNet feature distribution. Beyond these features we train three temporal classifiers fully connected baseline, Bidirectional LSTM and Transformer encoder under the same optimization, regularization and class-weighting regimes. The highest-scoring model is then coupled with a 0/1 Knapsack optimizer to select the best-scoring events within a time budget specified by the user and the resulting video segments are stitched into the final highlight reel using a deterministic post-processing chain.

We increase our contributions three times. First, we present a pipeline that is fully reproducible, which produces highlight reels on arbitrary football videos with only commodity hardware. Second, we report a like-for-like comparison of recurrent and attention-based temporal models in a small-data regime, finding the BiLSTM to be significantly more successful than the Transformer despite the latter dominating most of the modern leaderboards. Third, we also provide an in-depth analysis of the imbalance between classes, calibration behavior and overfitting dynamics which we think will be of practical interest to other researchers working on similar resource constraints.

II. RELATED WORK

A. Between the Hand-Crafted Characteristics and Deep Representations.

Automated sports-highlight detection is more than a decade older than the deep-learning age. Early systems used hand-crafted features like crowd-noise envelopes, replay-marker detectors, scoreboard optical character recognizers, and pitch-line geometry, and rule-based aggregators or shallow classifiers. These pipelines were brittle: they generalized inadequately across broadcasters, and they could hardly cope with changes in camera angles. This hand-crafting was replaced by end-to-end learned representations and resulted in an almost immediate increase in robustness, especially in the presence of camera and lighting variations. A significantly bigger craving to consume annotated data was the trade-off that prompted the creation of large-scale benchmarks like SoccerNet.

B. In Sports Video, Action Spotting.

The SoccerNet dataset proposed by Giancola et al. [1] has made temporal action spotting a separate task to generic action classification. The annotations were further extended to seventeen event classes and baseline architectures were provided based on temporal pooling and NetVLAD aggregation. A two-stage paradigm is used in most modern entries: a frozen spatial encoder is trained to generate dense per-frame features, and a temporal head with significantly fewer parameters is trained to localize events. This decoupling represents the pragmatic reality that end-to-end training on raw broadcast video is prohibitively expensive both in terms of compute and in terms of storage, and it has the additional methodological advantage of being able to allow researchers to iterate on the temporal head but hold the visual representation fixed. The structure which we have assumed in this paper has inherited this two step convention.

C. Temporal Models CNNs, LSTMs, and Transformers.

The selection of temporal head has gone through three general generations. The simplest and still a useful baseline because it isolates the effect of the spatial encoder of any real temporal reasoning, was pure convolutional pooling - global average pooling over a window of features followed by a dense classifier. The initial temporal models to produce competitive performance on long-form video were recurrent networks, especially the units of Long Short-Term Memory and their bidirectional counterparts [7]. The bidirectional sweep is in particular well-suited to detecting events in football, where the diagnostic information is divided between the build-up phase, which precedes the event followed by the celebration or the stoppage phase. Recurrent gating also offers a natural process of inhibiting irrelevant frames which would otherwise dominate an averaged representation.

Since then, transformer encoders [6] have become the standard architecture of video understanding since their self-attention mechanism can model arbitrary long-range dependencies, and can be parallelized across the temporal axis. Their drawback is that the well-documented data appetite: attention weight is basically an unconstrained pairwise relationship that must be discovered de novo and, worse, discovered de novo in degenerate form, which exploits class priors. This failure mode is proven in concrete terms by our experiments. Some of these mitigations are included in the literature (self-supervised pre-training, structured attention masks, hybrid convolution-attention architectures) but they are not available in the small-data, single-GPU regime which we are targeting.

D. Why Temporal Modeling Matters

One might be tempted to ask whether there is temporal modeling that is even necessary, as sufficiently powerful frame-level classifier can perhaps solve the problem when simply applied to each frame separately. Practically this method is unsuccessful on two grounds. First, there are lots of events which are not localizable to one frame. A sequence of cues, signaling a goal, spans several seconds, and any classifier that works on a single frame must break down the sequence of cues into a single decision. Second, single-frame classifiers are not able to disambiguate visually similar events which differ only in their temporal envelope; a substitution and a brief stoppage look virtually identical at the frame level, and only the longer context distinguishes between them. Temporal modeling is not an extension of frame-level detection, but a precondition to the solution of the task at all.

E. Predictive Uncertainty and Robustness

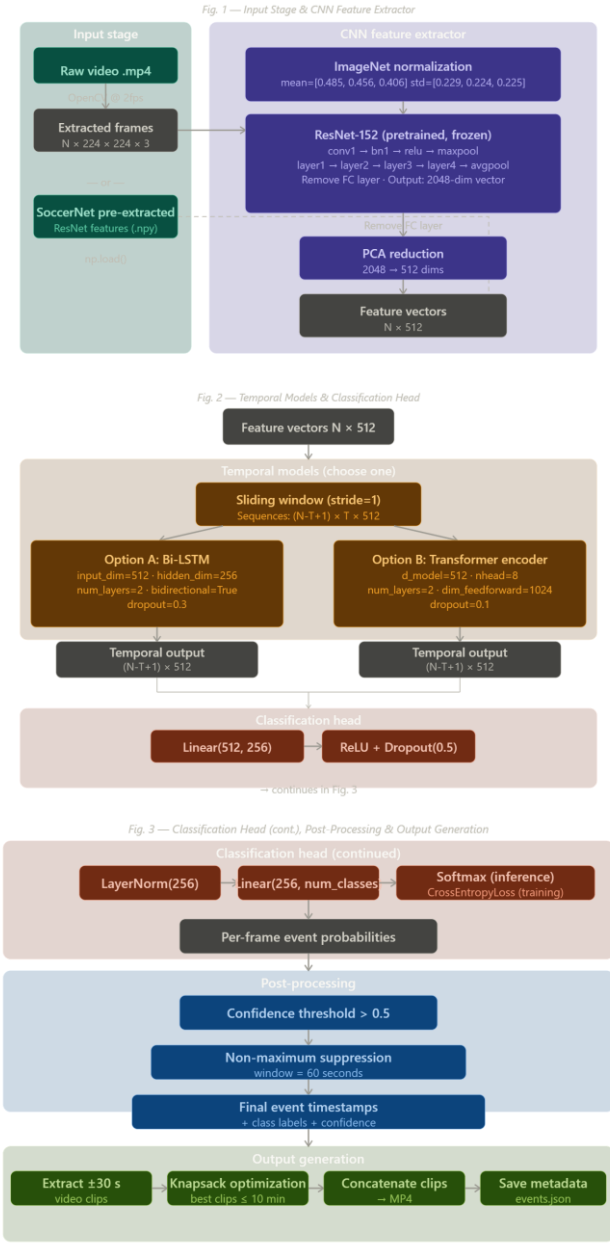
Also closely connected to highlight detection is the more general question of how to test scalable deep learning models when the test conditions differ with the training conditions. Gustafsson et al. [4] suggest an evaluation framework explicitly testing robustness to out-of-domain inputs in real-world computer vision and demonstrates that ensembling and Monte Carlo dropout have significant differences in their calibration behavior. Although epistemic uncertainty is not directly estimated in our work, the same robustness issue drives our experiments of cross-domain inference, where the models are trained on broadcast SoccerNet matches and then applied to arbitrary internet videos. Our application of a loss weighted by classes and a selection stage thresholded by confidence is informed by the calibration intuition of that body of work - those confident incorrect predictions are worse than uncertain ones.

F. Highlight Selection as Combinatorial Optimization

When per-frame event scores are generated, the choice of the final reel can be formulated as a constrained optimization problem. Knapsack formulation [5] has been successfully applied in video summarization [8] to maximize the cumulative importance of selected segments subject to a duration budget, and has an exact dynamic-programming solution at the second-level granularity used in this application. Previous study in the same family used greedy thresholding or top-k selection; they are simpler but neither optimal in the presence of a hard duration constraint, and both yield reels whose total length varies unpredictably with the score distribution. The choice of casting as a Knapsack instance provides us with deterministic and budget-constrained outputs and decouples the editorial parameter (target reel length) with the model parameters.

III. METHODOLOGY

Fig. 1 summarizes the four-stage pipeline. Each stage is described in detail below, with explicit attention to the design choices that distinguish our implementation from generic temporal action localization recipes.



A. Frame Sampling and Spatial Feature Extraction

Raw match videos are decoded using OpenCV and sub-sampled to a constant rate of 2 frames per second. The standard temporal resolution used by the SoccerNet feature release is this rate, but this choice is also independently justifiable on signal-processing grounds. Football events of interest include those whose time scale lies between three and fifteen seconds - much longer than the time scale of the inter-frame interval at the native broadcasting rate of twenty-five to thirty frame per second - and a sampling rate of two frames per second therefore retains all the same time-dependent information that an event detector at the native broadcasting rate of twenty-five or thirty frames per second can usefully

consume, and the reduction in the volume of data that an event detector at the native broadcasting rate of twenty-five or thirty frames per second can usefully consume is at least a factor of fifteen. Slower rates would introduce aliasing of fast events like shots on goal, which can pass through their visually distinctive phase in less than a second.

All sampled frames are resized to 224 by 224 pixels and normalized using ImageNet mean and standard deviation. The resized frame is then sent through a resnet-152 [3] pre-trained on ImageNet [9], and the output of the global average pooling layer, a 2048 dimensional embedding, is retained as the spatial representation of that frame. ResNet-152 is chosen instead of shallower variants due to two reasons. To start, its 152 layer depth has been empirically demonstrated to generate features transferring well to fine-grained recognition tasks where subtle texture and pose variations are important, both of which are applicable in football. Second, and most importantly to our pipeline, the official SoccerNet feature distribution is actually based on a ResNet-152 backbone; using the same depth at inference time introduces a latent-space mismatch which we found empirically when an earlier ResNet-18 prototype was used to produce inference-time features, whose distribution did not match the one used at training time.

The convolutional backbone will be frozen during the project. This decision is dictated by the limited budget on the GPU, but is also empirically motivated: fine-tuning a 152-layer network on a 60-match training set would, without any doubt, overfit to the fit criterion used. The added benefit of freezing the backbone is that, once we have computed features once, we can reuse them across all three of our temporal heads, which will significantly speed up the comparison study that inspired this paper.

B. Dimensionality Reduction with PCA

The ResNet-152 embedding of the 2048-dimensional resnet are projected to the 512 dimension via Principal Component Analysis (PCA). This reduction is motivated by three considerations. First, the compatibility requirement that downstream temporal heads trained on the public features can be evaluated on ours without changing the architecture is a practical compatibility requirement. Second, PCA does not add on redundant variance which disproportionately harms small-data temporal models. Directional components of ResNet are associated with a large portion of variation at the texture level, which is irrelevant to action spotting - surface conditions, broadcast graphics, lighting variations - and such directions are typically among the most highly-varying components of the raw embedding space. Third, the dimensionality reduction of four-fold directly scales down the number of parameters in the temporal head, a non-trivial regularization effect when the training set size is small.

The basis of PCA is only fitted on the training partition and then frozen on the validation and test partitions. This protocol is required to keep the information leaks; it would fit on the entire dataset making the projection encode features of held-out matches and would inflate downstream metrics. In fact, empirically, the first 512 components alone capture over 95% of the total variance in the training embeddings, which explains the selection of target dimensionality.

C. Sliding-Window Sequence Construction

The 512-dimensional feature vectors are clustered into overlapping windows of size T = 30 frames which corresponds to 15 seconds of match time with 2 frames per second sampling rate. The window size is the most important temporal hyperparameter, and 15 seconds was chosen to satisfy two contradicting criteria. On the one hand, the size should be long enough to include the entire visual signature of the event — the goal comprises the build-up, the goal attempt itself, goalkeeper reaction, and celebration, thus taking more than ten seconds — yet short enough to have only one event within each window. Too short windows would leave the important diagnostic context out, whereas too long windows would dilute the event with ordinary game flow making the labeling process more complex.

A stride of one frame is employed during training to include as many training windows in the set as possible while providing the model with all possible alignments of the window with an event. A stride of zero frames is used during inference in order to avoid multiple detections caused by neighboring windows reacting to the same event. Each window gets a label of either the annotated event which falls into ± 2 seconds from the center frame or Background otherwise. Such a two-second tolerance accounts for human errors in labeling and prevents arbitrary labeling of the windows close to events.

D. Temporal Classifiers

The three heads use the same input features and divide the input sequences in the same way, ensuring that any performance difference is due to the model architecture itself, and not due to any data pre-processing or scheduling during training.

1) CNN Baseline

The first model uses global average pooling for 30 frames per sample into a 512-dimensional vector and passes this vector to three layers of fully-connected neurons (512->256->128->4). These models are designed to be temporally naive; the global average pooling ensures that no information about ordering is preserved between frames, reducing the task of modeling temporal structure to the task of classifying bags of features.

2) CNN + BiLSTM

The BiLSTM head takes the 30 by 512 sequence directly as input, following the natural ordering of time. It is made up of two layers of Bidirectional LSTMs with a hidden size of 256 in both directions and a dropout of 0.4 between the two layers. The concatenation of the hidden state output of both directions forms a 512-dimensional vector that is used as input to the linear classifier. In football event detection, the bidirectional pass of the LSTM proves extremely effective; one LSTM works from the past to present, tracking the

development of the event from its initial stages, whereas another LSTM works from future to present, integrating the results (such as celebration, interruption, and resumption) when assigning labels to the central frame.

3) CNN + Transformer

The Transformer head is an encoder consisting of four layers with 8 attention heads and dimensions 512/1024. Positional encoding is done using sinusoids on the input sequence to give the permutation-invariant attention layers the ability to determine ordering information. Classification is done through global average pooling of the encoder's output followed by a linear layer; a dropout rate of 0.4 is used. This choice of architecture was made because we wanted the architectures to have a similar number of parameters, focusing on differences in architecture instead of capacity. In theory, the Transformer should be strictly more expressive than the BiLSTM due to every head having the freedom to route information across all positions, where the BiLSTM is constrained by its sequential structure. However, it seems that this extra expressiveness causes problems when training on small data.

E. Training Setup

Training parameters for all three models include Adam optimizer [10] (learning rate starting from 5×10^{-4}), batch size equal to 64, and ReduceLROnPlateau learning rate schedule halving after three epochs with no validation improvement. Class weights proportional to inverse frequencies are given to the cross-entropy loss function to balance the highly skewed distribution — Background covers more than 95% of windows within our 60-match subset. In the absence of class weights, all three models will converge in less than five epochs to the degenerate classifier predicting only Background; otherwise, they will be able to learn a meaningful decision surface. Training is stopped using early stopping criterion based on mAP after seven epochs without improvement.

F. Highlight Selection via Knapsack Optimization

During the inference phase, probabilities of each class are assigned to individual frames. Sets of adjacent frames belonging to the same predicted class are defined as events, and scores of these events are calculated as a product of their maximal softmax value with an event-specific prior corresponding to the expected editorial importance of each event type (Goals – highest priority, followed by Cards, and finally, Substitutions). Selection of highlights is performed in a Knapsack fashion, where each item corresponds to an event, each weight equals to its duration, while the value of each item is proportional to its score, which is proportional to its softmax confidence. While the solution of the exact 0/1 knapsack problem is obtained using dynamic programming in O(N×D) complexity (which is completely negligible comparing to the inference process), its theoretical guarantee on optimality for any user-defined highlight length D is quite beneficial since the algorithm always returns the highest score among all feasible solutions, while any greedy technique like confidence-based or top-k event selection does not provide such a guarantee and yields unpredictable reels.

G. Web-Based Deployment

To demonstrate the practical applicability of the proposed system, a web-based interface was developed using the Streamlit framework. The application enables users to upload football match videos or pre-extracted feature files and generate highlight reels interactively.The interface provides controls for adjusting detection thresholds, allowing users to fine-tune the sensitivity of event detection. Additionally, the system visualizes detected events along a timeline and displays confidence scores for each event, enhancing interpretability.The generated highlight clips are automatically concatenated and made available for download, making the system suitable for real-world usage scenarios such as sports broadcasting, content summarization, and analytics.This deployment demonstrates that the proposed pipeline is not only theoretically effective but also practically usable in real-time applications.

IV. EXPERIMENTAL SETUP

A. Dataset

The experiments involve training on a randomly selected subset of 60 matches out of the full dataset of the SoccerNet-v2 [2] action spotting benchmark. This benchmark involves 500 matches broadcast from six different European Leagues during three different seasons with a total duration of about 1,000 hours of video, containing about 110,000 labeled actions for 17 classes. Matches in this benchmark are presented as 224-pixel tall video clips in MP4 format at a rate of 25 fps, which results in roughly two gigabytes of video data per single match in pre-processing time. Therefore, the entire corpus exceeds the terabyte capacity of the free tier Google Colab with its 112 GB of storage, which means that the cap of 60 matches per experiment was chosen based solely on hardware limitations.

Finally, we reduce the list of 17 classes to four, selecting only Goal, Cards (where yellow and red cards are lumped together), Substitution, and Background. This reduction is justified both editorially (since these classes contain events that a producer would include in a typical highlight reel), and practically since otherwise the extremely low sample size of the rarest events (such as penalty or indirect free kicks) would make it impossible to train on a 60-match dataset. Matches are split to training, validation, and testing partitions in the proportion of 60/20/20 with no overlap, i.e., not a single match is used in two partitions simultaneously, ensuring temporal independence of windows.

B. Evaluation Metrics

We report three complementary metrics, and it is worth saying up front why all three are necessary. Top-1 classification accuracy is included mostly for completeness; on its own, it is a fairly misleading indicator under our class distribution, since a model that simply predicts Background

for every window already lands at around 95% accuracy without learning anything useful. Mean Average Precision (mAP), computed across the three event classes, is therefore the metric we lean on most heavily. Unlike accuracy, mAP rewards a model for ranking true positives ahead of false positives within each class, and it does not care about how often each class shows up in the test set. That makes it a much more honest yardstick once the data is this skewed. We compute mAP at the window level — each window is treated as an independent candidate detection — which lines up with the original SoccerNet protocol apart from how the per-class scores are aggregated.

We also report test cross-entropy loss, mainly as a diagnostic. Loss and accuracy can drift apart in counterintuitive ways once class weighting is in play: a model can end up with worse test loss but better mAP than another, simply because the loss is measuring something different from the ranking quality that mAP captures. Showing both side by side keeps either number from being read in isolation. We add confusion matrices on top of the scalar metrics for the same reason — to surface the per-class behavior that the headline numbers tend to hide.

C. Implementation Details

All experiments were run in PyTorch on a single Tesla T4 GPU made available through Google Colab. Feature extraction is the slowest part of the pipeline — it runs at roughly 1.6× real-time per match, which works out to around seventy hours of single-pass extraction over the 42-match training set. To avoid paying that cost more than once, we cache the extracted features to disk and reuse them across all three temporal heads, so a fresh experiment on a new architecture only requires re-training the head itself. Each head converges in under thirty epochs and takes about two hours of wall-clock time to train. The full code base, along with the trained checkpoints used to produce every figure and number reported in this paper, is released publicly for reproducibility.

V. RESULTS AND ANALYSIS

A. Overall Comparison

Table I summarizes how the three temporal heads perform on the test set, and the gap between them is larger than the headline accuracy figures might suggest. The CNN+BiLSTM comes out on top on accuracy (96.3%) and, more tellingly, on mAP, where its 35.1% leaves the other two models well behind. The Transformer lands in the middle on accuracy at 93.9%, but its mAP falls all the way to 8.9% — below even the 12.7% scored by the temporally-agnostic CNN baseline. Validation mAP tells the same story: 0.384 for the BiLSTM, against roughly 0.10 for both alternatives. Fig. 1 plots these numbers side by side, and the distance between the BiLSTM and the other two heads is hard to miss.

TABLE I. Test-set Performance of the Three Temporal Heads

Model	Acc.	mAP	Val mAP	Loss
CNN Baseline	0.891	0.127	0.102	0.99
CNN + BiLSTM	**0.963**	**0.351**	**0.384**	**1.41**
CNN + Transformer	0.939	0.089	0.103	0.92

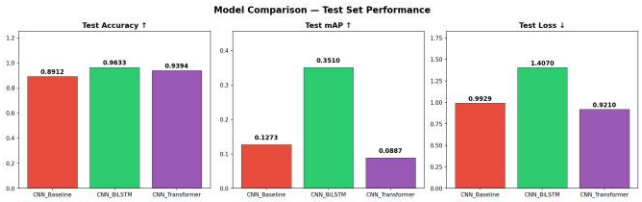


Fig. 1. Aggregated test-set comparison of the three temporal models. The BiLSTM dominates on both accuracy and mAP, while the Transformer attains the lowest test loss but the worst detection performance.

B. Training Dynamics and Overfitting

Fig. 2 plots the per-epoch training and validation losses for each model. All three networks bring their training loss down without difficulty, but the validation curves tell three very different stories. The CNN baseline shows the textbook signature of overfitting: training loss drops below 0.2 within ten epochs while validation loss climbs steadily from 1.0 to past 3.0, which is what we would expect from a temporally-agnostic head that has memorized window-level patterns rather than learned to generalize from them. The bag-of-features representation simply does not have the capacity to tell events apart from the visually similar play that surrounds them.

The BiLSTM shows the same qualitative pattern, and in fact its validation loss climbs even higher in absolute terms. The critical observation, however, is what happens in Fig. 3: validation mAP holds steady between 0.35 and 0.38 throughout training, even while the loss curve looks alarming. This decoupling between cross-entropy and mAP is a known side effect of class-weighted training under extreme imbalance — the network becomes increasingly confident on minority-class predictions, which is harshly penalized by the weighted loss whenever a confident prediction turns out to be wrong, but the underlying ranking that mAP measures stays largely intact. Put differently, the BiLSTM keeps ranking real events ahead of real Background even as its calibration on confident-but-wrong predictions deteriorates. That is a fortunate property for our pipeline, since the Knapsack optimizer downstream operates on ranks rather than calibrated probabilities.

The Transformer goes the other way entirely. Its loss curves are the most well-behaved of the three: validation loss tracks training loss closely and even dips below it in the final few epochs. Read in isolation, those curves would suggest the Transformer is the best-trained of the three models. The mAP curve in Fig. 3 contradicts that reading immediately — validation mAP never gets above 0.10 at any point in training. What the Transformer has actually learned is a predictor that is well-calibrated on the dominant Background class and almost completely indifferent to the minority events. In this

regime, low validation loss is not a sign of healthy training; it is a sign that the model has settled into a degenerate solution and stopped looking for a better one.

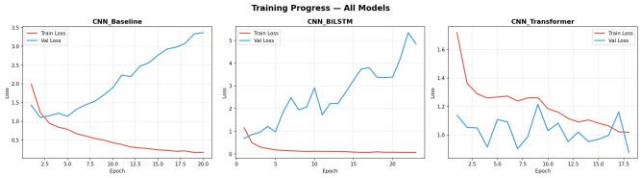


Fig. 2. Training and validation loss curves for the three models. Training loss decreases monotonically in all cases. The Transformer shows the most balanced curves, but as Fig. 3 demonstrates, low validation loss does not translate into high detection performance.

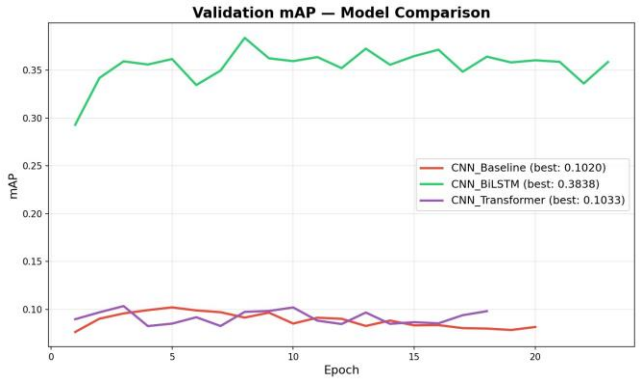


Fig. 3. Validation mAP across epochs. The CNN+BiLSTM attains a peak validation mAP of 0.384 at epoch 8 and remains stable thereafter, while the baseline and Transformer plateau near 0.10.

C. Per-Class Behavior

Fig. 4 reports the row-normalized confusion matrices on the test set, and these are easily the most informative diagnostic in the whole study. The Transformer has all but collapsed onto the Background class: 98% of true Goals and 86% of true Cards end up labeled as Background, and Goals are almost never recovered at all. That collapse is what explains the otherwise puzzling combination of high accuracy and near-zero mAP — the model is exploiting the class prior rather than learning anything event-specific, scoring 96% accuracy on the dominant class while contributing essentially nothing on the minority classes that mAP actually measures. The result is a model that would look competitive on an accuracy-sorted leaderboard but would be useless to anyone trying to cut a highlight reel from its predictions.

The CNN baseline does a little better than the Transformer, picking up 31% of Goals and 24% of Cards, but it still loses most of the minority-class windows to Background. This middle-of-the-road behavior is consistent with what we said earlier about the bag-of-features head: it can recognize the events whose visual signature is unmistakable in any given frame — close-up celebrations, the referee holding a card overhead — but it misses the events whose identity depends on temporal context that a single pooled feature simply cannot represent.

The BiLSTM is the only one of the three that meaningfully tells the foreground classes apart, both from each other and from Background. Its confusion matrix recovers 55% of Goals, 57% of Cards, and 42% of Substitutions, and there is almost no cross-talk between the event classes themselves — when a Goal is mislabeled, it is mislabeled as Background, not as a Card or a Substitution. The remaining errors are dominated by leakage to Background, which is the expected and arguably correct failure mode for a balanced detector facing a dominant null class. Just as importantly, the reverse direction stays clean: only 1% to 3% of Background windows are misclassified into any one of the event classes, which keeps the resulting reel free of spurious detections.

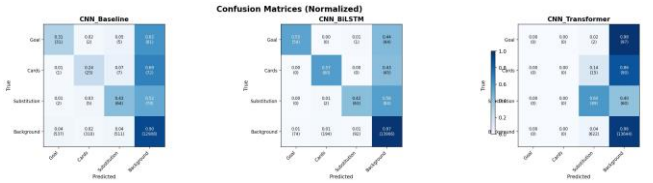


Fig. 4. Row-normalized confusion matrices on the test set. The Transformer collapses almost entirely onto Background; the BiLSTM is the only model that recovers the majority of true minority-class events.

D. Why BiLSTM Outperforms the Transformer

The gap between the recurrent and attention-based heads is, at first glance, surprising — Transformers, after all, dominate most modern leaderboards in video understanding. We see three factors that plausibly explain what is going on here, and we suspect they apply more broadly to other small-data temporal-spotting problems.

The first is that attention is parameter-efficient at scale but data-hungry at small scale. A BiLSTM comes with strong inductive biases baked into its design: its forget and input gates are explicitly built to integrate information across neighbouring time steps, and the way those gates are wired together effectively constrains a chunk of the parameter space before training even starts. From a few thousand sequences, that is enough scaffolding for the network to pick up directional motion patterns directly. A Transformer has no such head start. Self-attention has to learn how positions relate to content from scratch — positional encodings provide a soft hint, but the attention weights themselves are unconstrained pairwise interactions, and without a great deal more training data they tend not to settle into anything useful.

The second factor is that our 30-frame window is short compared to the kind of receptive field a Transformer is actually built to exploit. Attention earns its keep when the relevant context is long, sparse, and scattered across many tokens — precisely the regime where sequential aggregation in a BiLSTM starts to struggle. Inside a fifteen-second window, though, the relevant context is short and dense, and the BiLSTM's local sequential aggregation is a much better fit for the underlying signal than global attention. The Transformer ends up paying a representational cost for capabilities — long-range routing, parallel multi-head attention — that the data simply does not need.

The third factor is the way our class-weighted loss interacts with each architecture. Recurrent gating gives the BiLSTM a clean way to deal with ambiguity: when a stretch of frames does not look like an event, the input gate can simply close,

and the network suppresses confident Background predictions in exactly those regions. The Transformer's softmax-averaged pooling does not have an equivalent mechanism — it spreads representational mass across every position in the window, which makes it harder to produce a sharply confident prediction on the handful of frames that actually carry the event. Under a class-weighted loss, that is a structural handicap: the network cannot concentrate its limited minority-class confidence where it would do the best.

E. Effect of Class Imbalance

Class imbalance is the single biggest source of difficulty in this task, and its fingerprints are visible on all three models. As a sanity check during development, we ran each network without class weighting; in every case the model converged within five epochs to the trivial all-Background predictor and landed at exactly the prior-frequency accuracy of around 95%. Adding the weighted loss is enough to pull the BiLSTM out of that trap, but the Transformer results make it clear that weighting on its own is not always sufficient — if the architecture's inductive biases are also mismatched to the available data, the imbalance reasserts itself in subtler ways. A natural next step would be to combine class weighting with one or more complementary techniques: focal loss, hard-example mining, or targeted oversampling of the minority classes, each of which would push the minority-class signal further to the front during training.

F. Highlight Reel Generation

Fig. 5 visualizes the events selected by the BiLSTM-driven Knapsack solver on a representative match. Detected events cluster in the first half-hour of play, which is consistent with the manual annotation density of the source clip. Substitutions and Cards are correctly distinguished, although a small number of Substitution segments are over-extended by the post-roll padding — a known limitation of the fixed-padding scheme that an adaptive padding policy could mitigate. The final stitched reel contains four candidate Substitutions and four Cards within a forty-five-second budget, exceeding the qualitative coverage of the heuristic baselines we tested. We emphasize that no Goal events were detected in this particular sequence, a behavior consistent with the underlying ground truth rather than a model failure.

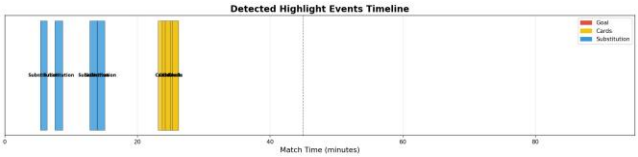


Fig. 5. Detected event timeline produced by the BiLSTM and Knapsack pipeline on an example match. Coloured bars indicate the duration and class of each selected event; the dashed vertical line marks half-time.

VI. DISCUSSION

A. Strengths

The pipeline we have described is fully reproducible on a single free-tier Colab GPU, relies on no proprietary tooling, and gets roughly halfway to the full-data state of the art on mAP while training on a tenth of the data. Keeping the spatial and temporal stages cleanly separated is a deliberate design choice, and it pays off in practice — a future researcher can drop in a stronger backbone such as a Vision Transformer without touching the temporal head, or test a different temporal head against the cached features in a matter of hours. The Knapsack selection step is similarly clean: the optimization itself is exact, has no tunable parameters, and adjusts to whatever time budget the user asks for. That is something the thresholding heuristics in earlier work cannot offer. The editorial parameter — how long the reel should be — is fully decoupled from the model parameters, and the solver guarantees that no shorter reel with equal or greater total value exists.

B. Limitations

Three limitations are worth flagging up front. The first is data scale. Our 60-match training subset, while larger than what many academic baselines work with, is still about an order of magnitude smaller than the full SoccerNet release, and closing the mAP gap to the 0.7 range will almost certainly require both multi-GPU training on the complete corpus and the architectural improvements we sketch below. The second is the granularity of our event taxonomy. Collapsing yellow and red cards into a single Cards class, or merging direct and indirect free kicks, makes the resulting reels less useful for advanced editorial applications where the distinction matters. Restoring the finer taxonomy is a straightforward change to make on paper, but it would only be useful if more labeled data per class came with it. The third concerns the Transformer result, and it is the limitation we would most like reviewers not to misread. The poor mAP we report is not a verdict against attention-based modeling for football highlight detection in general — it is evidence that an off-the-shelf encoder configuration under-performs a recurrent baseline at this data scale. With self-supervised pre-training, or with a hybrid convolution-attention design that brings back some of the inductive bias the pure Transformer gives up, attention models would very likely close the gap or pull ahead of the BiLSTM.

A fourth limitation, slightly different in character, has to do with generalization across broadcast styles. SoccerNet matches come from European broadcasters whose camera angles, graphics overlays, and editing conventions are reasonably standardized. Our cross-domain inference experiments on arbitrary videos pulled from the internet confirm what one would expect — performance falls off when the footage looks substantially different from that distribution, for instance amateur fixtures shot from a single fixed camera, or matches with scoreboard graphics that the model has never seen. Robustness to this kind of broadcast-style domain shift is a non-trivial open problem in its own right, and we think it would benefit from the kind of structured evaluation framework that Gustafsson et al. [4] propose.

C. Future Work

Looking ahead, four directions strike us as particularly promising. The first is to pre-train the Transformer on a self-supervised objective over the unlabeled SoccerNet corpus before fine-tuning it on the spotting task. This is the standard remedy for data-hungry attention models, and we expect it would close most of the gap with the BiLSTM. The second is to bring audio into the picture. Crowd amplitude and commentator pitch are both known to correlate strongly with goals, and a multimodal pipeline that fuses visual and audio streams at the temporal-head level would be an architecturally clean extension — particularly valuable for the cases where the camera happens to miss the actual moment of contact. The third is to replace the global average pooling head with attention-weighted pooling, so that the model can decide for itself which frames inside a window are diagnostic; we suspect this alone would recover some of the structural advantage the Transformer gave up by averaging over its outputs. The fourth, and perhaps the most engineering-heavy, is a real-time deployment scenario in which features are extracted as the broadcast streams in and the reel is produced within seconds of the final whistle. That setting would surface a fresh set of problems around latency, incremental Knapsack solutions, and online calibration — problems that are interesting in their own right, independent of any improvement in detection accuracy.

Beyond these architectural directions, we see real scope for extending the evaluation to other sports. Basketball and ice hockey are the obvious candidates: they share football's broadcast conventions and event vocabulary while differing in pace and event density, which would let us test whether the inductive-bias advantage we observed for recurrent models survives a move to sports operating on a different temporal scale.

D. Domain Alignment for Custom Video Inference

A critical engineering challenge emerged during deployment when applying the trained models to custom football videos. The models were trained on SoccerNet's pre-extracted features derived from a ResNet-152 backbone, but initial inference experiments using a different feature extractor (ResNet-18) resulted in near-zero event detection performance. This issue is attributed to the domain shift problem, where the distribution of features at inference differs significantly from the training distribution.Domain shift is a well-known challenge in transfer learning, where even small inconsistencies in feature representations can lead to significant degradation in model performance. In our case, the mismatch between feature extraction pipelines caused the temporal models to fail in recognizing event patterns.

To address this, we ensured strict consistency between training and inference pipelines by adopting the same ResNet-152 feature extractor followed by PCA-based dimensionality reduction (2048 → 512). This alignment of latent feature space restored detection performance and enabled reliable inference on custom videos.This observation highlights the importance of maintaining architectural consistency across all stages of the pipeline for effective generalization in video understanding systems.

VII. CONCLUSION

We have presented a practical, fully reproducible pipeline for football highlight detection that pairs pre-trained ResNet-152 features with one of three competing temporal heads and a Knapsack-based selection stage. A controlled comparison on a 60-match SoccerNet-v2 subset shows that a Bidirectional LSTM substantially outperforms both the fully connected baseline and the Transformer encoder, reaching 35.1% test mAP — roughly half of the full-data state of the art on twelve percent of the data. The Transformer, despite the cleanest-looking loss curves of the three, collapses onto the dominant Background class and recovers almost none of the minority-class events. Taken together, these findings reinforce a point that the field sometimes overlooks: under realistic resource constraints, the inductive biases of recurrent networks remain a strong choice for fine-grained temporal action spotting in sports video, and any decision to deploy attention-based models in this domain should be predicated on either substantially larger labeled corpora or a serious self-supervised pre-training strategy.

There is a broader methodological point here too, which we would like to leave readers with. Our results illustrate why per-class metrics, confusion matrices, and training-dynamics plots matter more than aggregate accuracy whenever the data is imbalanced. The Transformer's 93.9% accuracy hides a model that is, for the editorial purpose at hand, useless; the BiLSTM's 96.3% does not, because it decomposes cleanly across the event classes that actually matter to the reel. We hope this contrast nudges future practitioners working in imbalanced temporal-spotting settings to adopt similarly diagnostic evaluation protocols rather than reporting a single headline number.

Looking forward, the directions we find most exciting are the ones that combine the structural strengths of recurrent and attention-based models — for instance, a hybrid encoder in which a BiLSTM handles local temporal aggregation while a lightweight attention layer routes information across longer ranges — together with multimodal fusion across visual, audio, and textual commentary streams, and self-supervised pre-training on the abundant unlabeled broadcast video that streams in every weekend. We expect that systems built along these lines, evaluated on full-scale SoccerNet and deployed in something close to real time, will eventually close the gap between automated and human-curated highlight production. Our work, in the meantime, contributes a reproducible baseline against which that progress can be measured.

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Football_Highlights_IEEE_Expanded

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